

Does Rollout Training Help a Small CNN Emulator of Two-Scale Lorenz-96? Skill, Stability and Climatological Variance

Ibis-69E

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/autoregressive-emulator-on-lorenz-96>

Abstract—Machine-learning weather models are trained one step ahead but used for long autoregressive rollouts, and rollout (multi-step) losses are a common remedy. We isolate this one design choice in a small, fully controlled testbed: a circular 1D CNN with 10657 parameters emulates the 36 slow variables of two-scale Lorenz-96 (fast variables hidden), trained with a K -step rollout loss, $K \in \{1, 2, 4, 8\}$, five seeds each, with persistence, climatology and a ridge-fitted linear stencil as reimplemented baselines. All CNNs beat the baselines by a wide margin in ACC lead time (about 4.0 time units against 0.37 for the linear stencil). Rollout training gives a small, monotone gain at medium range: RMSE at lead 2 time units falls from 1.308 ($K=1$) to 1.243 ($K=8$), i.e. $K=1$ is 5.2% higher than $K=8$ (paired $p = 0.002$; $K=4$ is only marginal), at the price of a worse one-step RMSE. We find no stability or variance benefit to report: every free run of 200 time units survives for every K , including one-step training, and the climatological variance ratio stays within about one percent of one for all rollout-length settings. At lead 5 time units the CNN’s RMSE exceeds that of climatology although its ACC is still positive. The result is mixed and at small scale: the expected instability of one-step training did not appear in this setting.

Index Terms—Lorenz-96, autoregressive emulator, rollout loss, anomaly correlation, long-run stability, climatological variance

I. Introduction

Data-driven weather models such as GraphCast, FourCastNet and Keisler’s graph network are autoregressive: a network learns to map the atmospheric state at time t to $t + \Delta$ and is applied repeatedly to produce a medium-range forecast [1]–[3]. Neural general circulation models are trained through differentiable rollouts of many steps [4]. Training on one step only leaves a mismatch with deployment: errors made at step k become inputs at step $k+1$, and a model that is accurate one step ahead can drift off the attractor or blow up in long runs, a failure mode that has motivated noise injection, the pushforward trick, learned refinement, and post-hoc correction [5]–[8]. Rollout (multi-step) losses are a common remedy. The

cost is a loss that is blind to the sharpness of the one-step map and a training cost that grows linearly with the rollout length.

How much of the benefit of rollout training is skill, how much is stability, and what does it do to climatological variance? Large systems bundle rollout fine-tuning with changes of architecture, data and resolution, so these effects are hard to separate. We study the question in the cheapest chaotic multiscale testbed that still has an unresolved-scale problem: the two-scale Lorenz-96 system (used as a testbed in the multi-scale learning studies of [9], [10]), where the emulator sees only the 36 slow variables and the 360 fast variables act as unobserved forcing.

We pose four falsifiable hypotheses, registered before running (Sec. IV): (H1) multi-step training lowers medium-range RMSE; (H2) one-step training yields less stable free runs; (H3) multi-step training lowers climatological variance (blurring); (H4) the emulators beat persistence, climatology and a linear stencil in ACC lead time. We follow the WeatherBench convention of reporting RMSE and ACC against persistence and climatology as a function of lead time [11], [12].

Contributions. (i) A controlled sweep of the rollout length K for one small CNN, five seeds per setting, with every system measured by the same evaluation code. (ii) A measured, mixed answer: a small, monotone medium-range gain from large K (clear only for $K=8$), a one-step accuracy cost, and no detectable effect on stability or variance. (iii) Two ablations that bound the result: input-noise training (harmful here, and the only intervention that visibly damps variance) and a $8\times$ smaller training set (no effect). The study is a small CPU study; Sec. VII lists what it cannot say.

II. Related Work

ML weather forecasting and benchmarks. WeatherBench and WeatherBench 2 standardise verification of data-driven forecasts with RMSE and ACC against persistence, climatology and physical models [11], [12]; we adopt the per-lead reporting and the two trivial baselines. GraphCast, FourCastNet and the graph network of Keisler

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are autoregressive emulators at global scale [1]–[3]; NeuralGCM couples a learned component to a differentiable dynamical core and is trained through rollouts [4]. To our knowledge their rollout-length choices are made at full scale together with other design changes, and none can be re-run here; our CNN is a stand-in that shares only the autoregressive structure.

Lorenz-96 as a testbed. Chattopadhyay et al. compare reservoir computing, feed-forward networks and LSTMs for forecasting the slow variables of a multi-scale Lorenz-96 system [9]; Gagne et al. learn a stochastic parameterisation of the fast variables with a GAN [10]. Our emulator is deterministic and acts on the slow variables only, so unresolved-scale effects enter as irreducible one-step error.

Stable long rollouts. Learned coarse turbulence models and neural PDE solvers use training noise and the push-forward trick to counter error accumulation [5], [6]; PDE-Refiner replaces a single deterministic step by iterative refinement so that low-amplitude, high-frequency modes are not lost [7]; Thermalizer corrects emulator states back toward the invariant distribution using a diffusion model [8]. We test the simplest member of this family, training noise, and a multi-step loss, and report that in our setting the unmodified one-step model needed neither.

Gap. We found no cheap, multi-seed experiment that reports skill (RMSE, ACC versus lead), survival of long free runs and the climatological variance ratio together as a function of K ; this paper provides one. We do not claim it is the first.

III. Method

A. Truth: two-scale Lorenz-96

With $K=36$ slow variables X_k and $J=10$ fast variables $Y_{j,k}$ per slow variable (cyclic indices),

$$\dot{X}_k = -X_{k-1}(X_{k-2} - X_{k+1}) - X_k + F - \frac{hc}{b} \sum_j Y_{j,k}, \quad (1)$$

$$\dot{Y}_{j,k} = -cb Y_{j+1,k}(Y_{j+2,k} - Y_{j-1,k}) - cY_{j,k} + \frac{hc}{b} X_k, \quad (2)$$

with $F=10$, $h=1$, $c=10$, $b=10$; the fast variables form one ring of 360 cyclically indexed values. The system is integrated in NumPy with RK4 at step 0.005 and sampled every $\Delta=0.05$ time units (10 RK4 steps) after discarding 400 spin-up snapshots. Only $x_t = X(t\Delta) \in \mathbb{R}^{36}$ is given to the emulators; the Y are never used, so the one-step map is not closed and carries irreducible error.

B. CNN emulator and the rollout loss

Let $z = (x - \mu)/\sigma$ with μ, σ the scalar mean and standard deviation of the training data. The emulator is a residual circular CNN,

$$\begin{aligned} \hat{z}_{t+1} &= \hat{z}_t + f_\theta(\hat{z}_t), \\ f_\theta &= \text{conv}_4 \circ \text{GELU} \circ \text{conv}_3 \circ \text{GELU} \\ &\quad \circ \text{conv}_2 \circ \text{GELU} \circ \text{conv}_1, \end{aligned} \quad (3)$$

with 4 circular 1D convolutions (kernel 5, channels $1 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1$). Starting from a training state z_t and

unrolling $\hat{z}_t^{(0)} = z_t$, $\hat{z}_t^{(k)} = \hat{z}_t^{(k-1)} + f_\theta(\hat{z}_t^{(k-1)})$, the K -step rollout loss is

$$\mathcal{L}_K(\theta) = \frac{1}{K} \sum_{k=1}^K \frac{1}{36} \left\| \hat{z}_t^{(k)} - z_{t+k} \right\|_2^2, \quad (4)$$

averaged over a minibatch of random windows, with full back-propagation through the unrolled steps. $K=1$ is ordinary one-step training. For the noise ablation the initial state is perturbed, $\hat{z}_t^{(0)} = z_t + \epsilon$, $\epsilon \sim \mathcal{N}(0, s^2 I)$, while targets stay clean; the number of optimiser steps, batch size and schedule are identical for every K (so larger K costs proportionally more compute per step).

C. Reimplemented baselines

Persistence: $\hat{x}_{t+\tau} = x_t$. Climatology: $\hat{x}_{t+\tau} = \mu$ (the training mean). Linear stencil (ridge): a translation-invariant linear one-step model $\hat{z}_{t+1,k} = z_{t,k} + \sum_{s=-2}^2 w_s z_{t,k+s} + w'_0$, with w the ridge solution (penalty 0.001) of the one-step increment regression over all training sites and times; it is the linear counterpart of the CNN with the same receptive field. All three use the same evaluation code and data as the CNNs. These are simple baselines chosen for the controlled question; they are not published ML forecasters.

D. Metrics

For forecast windows w and lead τ (in steps), $\text{RMSE}(\tau) = \sqrt{\text{mean}_{w,k}(\hat{x} - x)^2}$ and

$$\text{ACC}(\tau) = \text{mean}_w \frac{\sum_k (\hat{x}_k - \mu)(x_k - \mu)}{[\sum_k (\hat{x}_k - \mu)^2 \sum_k (x_k - \mu)^2]^{1/2}}, \quad (5)$$

the pattern correlation of anomalies about the training climatology (set to 0 for the constant climatology forecast, whose anomaly is zero). The ACC lead time is the lead at which mean ACC first falls below 0.6 (linear interpolation; capped at the horizon). In a free run of 200 time units from 16 test states, a member diverges if any value is non-finite or $|x| > 50$ (the truth stays below about 15); stable_frac is the surviving fraction and survival_time the mean survival time. For surviving members, after dropping the first 10 time units, we report the variance ratio var_ratio = $\text{Var}_{\text{emu}}/\text{Var}_{\text{truth}}$ of all pooled values, the roughness ratio (the same ratio for the discrete Laplacian $x_{k+1} - 2x_k + x_{k-1}$, which is sensitive to smoothing and to grid-scale noise) and the Kolmogorov–Smirnov distance ks between pooled emulator and truth marginals; truth is every test-trajectory state. Persistence and climatology have no free run (undefined or trivial), so they are omitted from long-run columns.

IV. Experimental Setup

Data. For each seed s we simulate independent trajectories: 32 training (1250 snapshots each), 8 validation (500) and 32 test (1500), all from different random initial conditions, so the splits are disjoint trajectories. The seed determines the data, the network initialisation and the

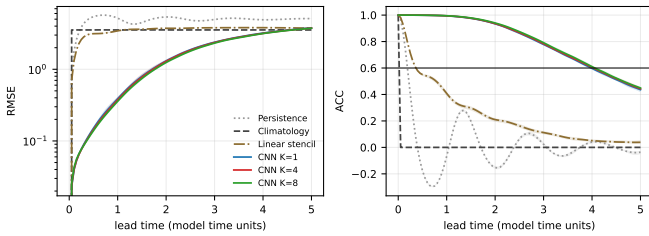


Fig. 1. RMSE (log scale, left) and ACC (right) against lead time; lines are seed means, bands the min-max over seeds. The three CNN curves nearly coincide. The horizontal line is the ACC=0.6 threshold.

minibatch sequence; the same data seed is shared by all systems, which is why we compare systems with paired tests over seeds. Forecasts start every 50th snapshot of each test trajectory (896 windows per seed) and run for 100 steps (5 time units); we report leads 0.5, 1, 2, 3 and 5 time units as t05...t50, and 1step for a single step.

Training and tuning. Every CNN is trained for 1500 Adam steps (batch 32, cosine decay, gradient-norm clip 1). The only tuned hyper-parameter is the peak learning rate, chosen from $\{0.001, 0.003\}$ for $K \in \{1, 4\}$ in 4 runs on one extra seed (100) by validation RMSE; 0.003 won for both on the validation RMSE at leads 1 and 2 and was used for every K (no per- K tuning, and no tuning on test). The selected rate is the upper edge of this two-point grid, so a larger rate was not tried and could change the ordering. Persistence and climatology have no parameters; the linear stencil is closed-form with a fixed ridge penalty. The tuning budget is small and $K=2, 8$ were not tuned separately, so the shared rate may favour $K=1, 4$; a longer-trained or larger network could also change the picture (Sec. VII).

Registered hypotheses and tests. Tests were fixed in proposal.md before the runs: paired comparison over 5 seeds (rh compare, paired t -test, $\alpha=0.05$, no multiplicity correction) of $K=4$ and $K=8$ against $K=1$ for H1 (RMSE at lead 2), stable fraction for H2, variance ratio for H3, ACC lead time of CNN against the baselines for H4. Groups: main (baselines, $K=1, 4, 8$), sweep_K ($K=2$), abl_noise and sweep_ntrain.

Compute. Everything ran on one shared CPU machine (macOS, 2 threads per process, at most 2 processes at once), each run within the 6-minute budget. The CNN has 10657 parameters. The code is in method/, the driver in experiments/run_all.sh, and every number in this paper is read from the run registry.

V. Results

Table I and Fig. 1 give the forecast skill of every system in the main group; Table III gives the free-run statistics. Values are means $\pm$ standard deviation over 5 seeds; differences and p -values come from rh compare (paired t -test over seeds, reference CNN $K=1$, so a positive Δ means lower RMSE than $K=1$).

A. H4: the emulators have skill far beyond the baselines

Persistence has an ACC lead time of only 0.19 time units, because the L96 waves propagate and persistence ACC oscillates around zero (Fig. 1); the linear stencil reaches 0.37, and the CNNs between 3.98 ($K=1$) and 4.05 ($K=8$). The one-step RMSE of $K=1$ is 0.023 against 0.698 for the linear stencil, which is hardly better than persistence’s 0.923, so the nonlinearity matters. H4 is supported: e.g. the lead-time difference between $K=1$ and persistence is 3.79 time units (paired $p = 2.526 \times 10^{-9}$).

Where it fails. At 5 time units the CNN RMSE (3.78 for $K=1$) is above climatology’s 3.54, although its ACC is still 0.44: the emulator is sharp (it does not blur toward the mean, see H3), and a plausible reading is that once phase errors grow a sharp forecast is penalised by RMSE more than the constant mean is (we did not test this with a blurred model). Judged by RMSE alone, the CNN beats climatology only before this crossing; at 3 time units it is still better (2.39 against 3.54). At 5 time units the linear stencil (3.75) is statistically indistinguishable from $K=1$ in RMSE (paired $p = 0.25$), and both are worse than climatology; the linear model reaches this level by decaying to the mean, not by being skilful.

B. H1: rollout training helps modestly at medium range

RMSE at lead 2 falls monotonically in the mean with K : 1.308 ($K=1$), 1.300 ($K=2$), 1.290 ($K=4$), 1.243 ($K=8$) (Fig. 2). Against $K=1$, $K=8$ improves by 0.065 ($K=1$ is 5.2% higher than $K=8$; paired $p = 0.002$, Cohen’s $d = 1.55$; Welch $p = 0.04$), and $K=4$ by 0.018 ($K=1$ is 1.4% higher than $K=4$; paired $p = 0.032$, but Welch $p = 0.49$). The same ordering holds at leads 1, 3 and 5 (Table I) and for the ACC lead time, which rises from 3.984 ($K=1$) to 4.053 ($K=8$; paired $p = 0.002$). H1 is therefore supported for $K=8$ and only weakly for $K=4$: the effect is small (about a twentieth of the error at $K=8$, below the relative gain we had set as a meaningful threshold in the study configuration), the $K=4$ evidence rests on the paired test alone, no multiplicity correction is applied, and seed-to-seed variation (std 0.034) is a sizeable fraction of the effect. At the longest lead 5, $K=4$ is not distinguishable from $K=1$ (paired $p = 0.07$), whereas $K=8$ is (paired $p = 0.003$), though the absolute gain is only 0.04.

Cost. The one-step RMSE gets worse with K (0.0229, 0.0230, 0.0237, 0.0253 for $K = 1, 2, 4, 8$; $K=8$ vs $K=1$ paired $p = 9.448 \times 10^{-7}$), and the training compute per step grows about linearly in K .

C. H2: no instability was observed

All free runs of 200 time units survive for the linear stencil and for every CNN: stable_frac is 1.0 for $K = 1$ and for every other K (Table III). H2 is refuted in this setting: there is nothing for multi-step training to fix, and rh compare has no variance to test. We only observed $K=1$ with $\Delta=0.05$; other steps, architectures or noisier data might behave differently.

TABLE I

Forecast skill, group main. RMSE at 1 step and at leads of 1, 2, 3 and 5 time units (t10, t20, t30, t50) and ACC lead time (time units until mean ACC < 0.6); lower RMSE and higher lead time are better. Bold/underline mark best/second best per column. Climatology’s ACC is zero by definition, so its “lead time” is an interpolation artefact.

Method	rmse_1step ↓	rmse_t10 ↓	rmse_t20 ↓	rmse_t30 ↓	rmse_t50 ↓	acc_leadtime ↑	n
Persistence	0.923 ± 0.005	5.160 ± 0.020	5.364 ± 0.014	4.948 ± 0.027	5.088 ± 0.031	0.190 ± 0.001	5
Climatology	3.538 ± 0.007	3.534 ± 0.008	3.540 ± 0.010	3.538 ± 0.006	3.536 ± 0.009	0.020 ± 0.000	5
Linear stencil (ridge)	0.698 ± 0.002	3.435 ± 0.036	3.732 ± 0.028	3.794 ± 0.021	3.753 ± 0.011	0.366 ± 0.009	5
CNN K=1	0.023 ± 0.000	0.388 ± 0.015	1.308 ± 0.034	2.385 ± 0.038	3.775 ± 0.041	3.984 ± 0.038	5
CNN K=4	<u>0.024 ± 0.000</u>	<u>0.374 ± 0.021</u>	<u>1.290 ± 0.044</u>	<u>2.359 ± 0.044</u>	3.752 ± 0.035	<u>4.030 ± 0.050</u>	5
CNN K=8	0.025 ± 0.000	0.357 ± 0.024	1.243 ± 0.048	2.317 ± 0.037	<u>3.733 ± 0.037</u>	4.053 ± 0.043	5

TABLE II

ACC at leads 0.5, 1, 2, 3, 5 time units (acc_t05...acc_t50), group main. The climatology forecast has zero anomaly, so its ACC is set to 0.

Method	acc_t05 ↑	acc_t10 ↑	acc_t20 ↑	acc_t30 ↑	acc_t50 ↑	n
Persistence	-0.176 ± 0.006	-0.067 ± 0.007	-0.153 ± 0.008	0.023 ± 0.010	-0.036 ± 0.010	5
Climatology	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	5
Linear stencil (ridge)	0.550 ± 0.006	0.408 ± 0.012	0.210 ± 0.011	0.112 ± 0.011	0.039 ± 0.010	5
CNN K=1	0.999 ± 0.000	0.994 ± 0.000	0.934 ± 0.003	0.778 ± 0.006	0.436 ± 0.012	5
CNN K=4	0.999 ± 0.000	0.995 ± 0.001	0.936 ± 0.004	0.783 ± 0.007	0.443 ± 0.011	5
CNN K=8	0.999 ± 0.000	0.995 ± 0.001	0.940 ± 0.004	0.791 ± 0.006	0.449 ± 0.009	5

TABLE III

Free-run statistics (200 time units from 16 states), group main. var_ratio, roughness_ratio: emulator/truth, target 1; the bold/underline ranking in this table is by “lower is better” and should be ignored for these two columns (the linear stencil, whose runs decay to the mean, is “best” by that rule). Persistence and climatology have no free run.

Method	stable_frac ↑	survival_time ↑	var_ratio ↓	roughness_ratio ↓	ks ↓	n
Linear stencil (ridge)	1.000 ± 0.000	200.000 ± 0.000	0.001 ± 0.000	0.001 ± 0.000	0.468 ± 0.002	5
CNN K=1	1.000 ± 0.000	200.000 ± 0.000	1.003 ± 0.003	1.005 ± 0.007	0.002 ± 0.001	5
CNN K=4	1.000 ± 0.000	200.000 ± 0.000	1.002 ± 0.004	1.005 ± 0.010	<u>0.002 ± 0.002</u>	5
CNN K=8	1.000 ± 0.000	200.000 ± 0.000	<u>0.998 ± 0.003</u>	<u>0.997 ± 0.006</u>	0.002 ± 0.001	5

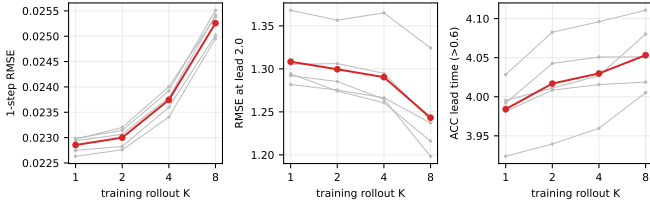


Fig. 2. Effect of the training rollout length K (groups main and sweep_K): one-step RMSE, RMSE at lead 2 and ACC lead time. Grey lines are the five seeds (data seeds are shared across K), red is the seed mean.

D. H3: climatological variance is preserved, not blurred

The variance ratio of the CNNs is 1.003 ($K=1$), 1.002 ($K=4$) and 0.998 ($K=8$); the roughness ratio is 1.005, 1.005 and 0.997, and the KS distance is at most 0.002 for $K=1$. The difference $K=8$ vs $K=1$ is 0.005 (paired $p = 0.08$, Welch $p = 0.030$) and $K=4$ vs $K=1$ is 0.001 ($p = 0.63$). H3 is not supported: the sign is the predicted one for $K=8$ but the effect is below one percent and not significant under the registered paired test (the Welch test alone gives a smaller p , but the effect is below one percent and we do not rest a claim on it), so we cannot claim multi-

step training reduces variance. By contrast the ridge linear stencil is “stable” only in the trivial sense: its free runs decay to the mean (variance ratio 0.0013, KS 0.47), which shows that the stability criterion must be read together with the variance ratio.

VI. Ablations and Analysis

We run three further studies; Table IV lists them. Reference values are the CNN $K=1$ and $K=4$ rows of Table I.

Intermediate rollout length ($K=2$). The $K=2$ model sits between $K=1$ and $K=4$ on every skill metric: RMSE at lead 2 is 1.300 and ACC lead time 4.017, with variance ratio 1.001. The improvement from $K=1$ to $K=4$ is thus gradual, and only the largest K separates clearly from noise (Fig. 2). We did not test $K > 8$ (cost) so we cannot say where gains saturate or reverse.

Input noise hurts here, and is the only intervention that damps variance. Training $K=1$ with input noise of std 0.03 raises RMSE at lead 2 to 1.564 (reference 1.308) and shortens the ACC lead time to 3.81; with std 0.1 the numbers are 2.770 and 2.47. The variance ratio falls to 0.985 and 0.902, the roughness ratio to 0.973 and 0.751, and the KS distance rises to 0.046 for the larger

TABLE IV

Sensitivity and ablation groups (mean $\pm$ std over 5 seeds). Top: rollout length $K=2$ (sweep_K). Middle: $K=1$ with Gaussian input noise of std 0.03 and 0.1 (abl_noise). Bottom: $K=1, 4$ trained on only 4 of the 32 training trajectories (sweep_ntrain). In each block the bold/underline ranking is only among that block’s rows; the reference $K=1, 4, 8$ rows are in Tables I and III.

Method	rmse_1step $\downarrow$	rmse_t20 $\downarrow$	rmse_t50 $\downarrow$	acc_leadtime $\uparrow$	var_ratio $\downarrow$	ks $\downarrow$	n
CNN $K=2$	0.023 ± 0.000	1.300 ± 0.034	3.762 ± 0.045	4.017 ± 0.053	1.001 ± 0.003	0.002 ± 0.001	5

Method	rmse_1step $\downarrow$	rmse_t20 $\downarrow$	rmse_t50 $\downarrow$	acc_leadtime $\uparrow$	var_ratio $\downarrow$	ks $\downarrow$	n
CNN $K=1$ noise=0.03	0.025 ± 0.000	1.564 ± 0.032	3.822 ± 0.024	3.808 ± 0.052	0.985 ± 0.007	0.004 ± 0.001	5
CNN $K=1$ noise=0.1	<u>0.086 ± 0.002</u>	<u>2.770 ± 0.045</u>	<u>4.230 ± 0.041</u>	<u>2.467 ± 0.041</u>	<u>0.902 ± 0.013</u>	<u>0.046 ± 0.009</u>	5

Method	rmse_1step $\downarrow$	rmse_t20 $\downarrow$	rmse_t50 $\downarrow$	acc_leadtime $\uparrow$	var_ratio $\downarrow$	ks $\downarrow$	n
CNN $K=1$ ntrain=4	0.023 ± 0.000	<u>1.320 ± 0.038</u>	<u>3.779 ± 0.038</u>	<u>3.978 ± 0.041</u>	<u>1.003 ± 0.003</u>	<u>0.002 ± 0.001</u>	5
CNN $K=4$ ntrain=4	<u>0.024 ± 0.000</u>	1.296 ± 0.027	3.752 ± 0.025	4.020 ± 0.029	1.002 ± 0.003	0.002 ± 0.001	5

noise. All noisy runs remain stable (stable_frac 1.0). Noise therefore damps small scales (the noisy model learns a smoothed map because it cannot rely on grid-scale detail of the input), which is consistent with the roughness drop; we did not isolate the mechanism further, so this is an interpretation. Noise would be expected to pay off only where the unperturbed model is unstable, which was not the case; the comparison is cross-group, so the differences are read from the table rather than from rh compare.

Training-set size. Reducing training data to 4 trajectories (about 5000 snapshots) leaves everything unchanged: RMSE at lead 2 is 1.320 for $K=1$ and 1.296 for $K=4$, against 1.308 and 1.290 with all 32; both stay stable with variance ratio 1.003 and 1.002. The ordering $K=4$ better than $K=1$ at this lead is preserved, with a similar small gap. The relevant limit is therefore not the amount of data but the information in the slow variables (unobserved fast scales) and the short training; at this data size we cannot say whether rollout training matters more at even smaller data.

Why is one-step training stable? We have no run that isolates a mechanism. Candidate explanations that we did not test include the very small one-step error (0.023 against a standard deviation of 3.54), the residual parametrisation and the dissipative attractor of Lorenz-96; they should be read as conjectures.

VII. Limitations

Scale and statistics. A single small architecture (10657 parameters), one learning rate shared across K (tuned for $K = 1, 4$ only), 1500 optimiser steps and 5 seeds. The $K=8$ vs $K=1$ effect is about five percent in RMSE at one lead with a paired p that is not corrected for the many metrics and leads we inspected; the $K=4$ effect is not robust to the unpaired test. Rollout lengths above 8 were not run, nor equal-compute comparisons (a $K=8$ step costs roughly eight times a $K=1$ step; a $K=1$ model trained eight times longer might close the gap).

Setting. A single system with one parameter set ($F=10$, $c=b=10$) and one sampling step. Instabilities of one-

step emulators are reported for harder, higher-dimensional problems; failing to see them here does not show they are absent elsewhere. Our stability criterion (divergence beyond $|x| > 50$ within 200 time units from 16 states) cannot detect slow drift of the climate that stays bounded; we tracked variance, roughness and a marginal KS distance, but not spectra, extremes or temporal correlations.

Baselines. Persistence, climatology and a linear stencil are reimplemented and simple; there is no comparison with a published emulator, a data-assimilation-style ensemble, or with the true one-scale system. The metrics are deterministic and say nothing about calibration. ACC is relative to a scalar climatology (the system is statistically homogeneous), and climatology’s ACC is defined as 0.

Evaluation. Test windows from the same trajectory overlap in the sense of being correlated; they are averaged before the seed-level tests, which treat seeds as the replicates. Seeds change data, initialisation and minibatches together, so we cannot attribute variance to any one of them. All results are for a CPU-scale model; a full study would add larger networks, longer training, other noise strategies (pushforward, refinement), equal-compute budgets, ensembles with CRPS, and multiple Lorenz-96 regimes.

VIII. Conclusion

In a controlled two-scale Lorenz-96 study with a small circular CNN, five seeds and reimplemented baselines, rollout training gave a modest, monotone improvement in medium-range skill that is clear only for $K=8$ (RMSE at lead 2: 1.308 to 1.243) and costs one-step accuracy. We found no stability problem for one-step training, no measurable variance loss from multi-step training, and a negative effect of input noise. All emulators beat persistence, climatology and a linear stencil in ACC lead time, but beyond about 4.5 time units the CNN RMSE exceeds that of climatology. The claims are limited to this small setting; a stability benefit of rollout training, if it exists, needs a harder regime than the one studied here.

References

- [1] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, G. Holland, O. Vinyals, J. Stott, A. Pritzel, S. Mohamed, and P. Battaglia, “GraphCast: Learning skillful medium-range global weather forecasting,” arXiv preprint arXiv:2212.12794, 2022. [Online]. Available: <https://arxiv.org/abs/2212.12794>
- [2] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Li, K. Azizzadenesheli, P. Hassanzadeh, K. Kashinath, and A. Anandkumar, “FourCastNet: A Global Data-driven High-resolution Weather Model using Adaptive Fourier Neural Operators,” arXiv preprint arXiv:2202.11214, 2022. [Online]. Available: <https://arxiv.org/abs/2202.11214>
- [3] R. Keisler, “Forecasting Global Weather with Graph Neural Networks,” arXiv preprint arXiv:2202.07575, 2022. [Online]. Available: <https://arxiv.org/abs/2202.07575>
- [4] D. Kochkov, J. Yuval, I. Langmore, P. Norgaard, J. Smith, G. Mooers, M. Klöwer, J. Lottes, S. Rasp, P. Düben, S. Hatfield, P. Battaglia, A. Sanchez-Gonzalez, M. Willson, M. P. Brenner, and S. Hoyer, “Neural General Circulation Models for Weather and Climate,” arXiv preprint arXiv:2311.07222, 2023. [Online]. Available: <https://arxiv.org/abs/2311.07222>
- [5] K. Stachenfeld, D. B. Fielding, D. Kochkov, M. Cranmer, T. Pfaff, J. Godwin, C. Cui, S. Ho, P. Battaglia, and A. Sanchez-Gonzalez, “Learned Coarse Models for Efficient Turbulence Simulation,” arXiv preprint arXiv:2112.15275, 2021. [Online]. Available: <https://arxiv.org/abs/2112.15275>
- [6] J. Brandstetter, D. Worrall, and M. Welling, “Message Passing Neural PDE Solvers,” arXiv preprint arXiv:2202.03376, 2022. [Online]. Available: <https://arxiv.org/abs/2202.03376>
- [7] P. Lippe, B. S. Veeling, P. Perdikaris, R. E. Turner, and J. Brandstetter, “PDE-Refiner: Achieving Accurate Long Rollouts with Neural PDE Solvers,” arXiv preprint arXiv:2308.05732, 2023. [Online]. Available: <https://arxiv.org/abs/2308.05732>
- [8] C. Pedersen, L. Zanna, and J. Bruna, “Thermalizer: Stable autoregressive neural emulation of spatiotemporal chaos,” arXiv preprint arXiv:2503.18731, 2025. [Online]. Available: <https://arxiv.org/abs/2503.18731>
- [9] A. Chattopadhyay, P. Hassanzadeh, and D. Subramanian, “Data-driven prediction of a multi-scale Lorenz ’96 chaotic system using deep learning methods: Reservoir computing, ANN, and RNN-LSTM,” arXiv preprint arXiv:1906.08829, 2019. [Online]. Available: <https://arxiv.org/abs/1906.08829>
- [10] D. J. Gagne, H. M. Christensen, A. C. Subramanian, and A. H. Monahan, “Machine Learning for Stochastic Parameterization: Generative Adversarial Networks in the Lorenz ’96 Model,” arXiv preprint arXiv:1909.04711, 2019. [Online]. Available: <https://arxiv.org/abs/1909.04711>
- [11] S. Rasp, P. D. Dueben, S. Scher, J. A. Weyn, S. Mouatadid, and N. Thuerey, “WeatherBench: A benchmark dataset for data-driven weather forecasting,” arXiv preprint arXiv:2002.00469, 2020. [Online]. Available: <https://arxiv.org/abs/2002.00469>
- [12] S. Rasp, S. Hoyer, A. Merose, I. Langmore, P. Battaglia, T. Russel, A. Sanchez-Gonzalez, V. Yang, R. Carver, S. Agrawal, M. Chantry, Z. B. Bouallegue, P. Dueben, C. Bromberg, J. Sisk, L. Barrington, A. Bell, and F. Sha, “WeatherBench 2: A benchmark for the next generation of data-driven global weather models,” arXiv preprint arXiv:2308.15560, 2023. [Online]. Available: <https://arxiv.org/abs/2308.15560>